

Supplementary Material: Information-Theoretic Analysis of LLM-Guided Tree Search with Empirical Validation

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1 Proof of Theorem 1 (Bayesian Regret Upper Bound)

1.1 Regret Decomposition

Let P^* be the true optimal path distribution and Q_{LLM} the LLM-derived prior. The cumulative Bayesian regret after T iterations decomposes as:

$$\mathbb{E}[\text{Regret}(T)] = \underbrace{\mathbb{E}\left[\sum_{t=1}^T (V^*(s_0) - V^{\pi_t}(s_0))\right]}_{\text{finite-sampling regret}} + \underbrace{\mathbb{E}\left[\sum_{t=1}^T (V^{\pi_t}(s_0) - V^Q(s_0))\right]}_{\text{prior misspecification regret}} \quad (1)$$

1.2 Finite-Sampling Term

By the information ratio analysis of Russo & Van Roy (2016) and its extension to tree search by Greshler et al. (2024, Theorem 1), the finite-sampling regret of Thompson Sampling with prior Q_{LLM} satisfies:

$$\mathbb{E}\left[\sum_{t=1}^T (V^*(s_0) - V^{\pi_t}(s_0))\right] \leq R_{\max} H \sqrt{|\mathcal{Z}|T \cdot \frac{H(P^*)}{2}} \quad (2)$$

where $H(P^*)$ is the Shannon entropy of the optimal path distribution and $|\mathcal{Z}| = K^H$ is the size of the solution space.

1.3 Prior Misspecification Term

Let $D_t = D_{\text{KL}}(P_t^* \parallel Q_{\text{LLM},t})$ be the KL divergence between the true posterior and the LLM-derived posterior at iteration t . By Lemma 1 (Posterior KL Decay), $D_t \leq D_0/(1 + t/A_0)$ where $D_0 = D_{\text{KL}}(P^* \parallel Q_{\text{LLM}})$ and $A_0 = \alpha_q + \beta_q$ is the Beta prior strength.

The cumulative misspecification regret satisfies:

$$\mathbb{E} \left[\sum_{t=1}^T (V^{\pi_t}(s_0) - V^Q(s_0)) \right] \leq R_{\max} H \sqrt{|\mathcal{Z}|T} \cdot \sqrt{8D_{\text{KL}}(P^* \parallel Q_{\text{LLM}})A_0} \quad (3)$$

This follows from Pinsker's inequality ($\|P - Q\|_1 \leq \sqrt{2D_{\text{KL}}(P \parallel Q)}$) and the cumulative KL sum $\sum_{t=1}^T D_t \leq D_0 A_0 \log(1 + T/A_0)$.

1.4 Combined Bound

Adding both terms yields the stated upper bound:

$$\mathbb{E}[\text{Regret}(T)] \leq R_{\max} H \sqrt{|\mathcal{Z}|T} \times \left(\sqrt{\frac{H(P^*)}{2}} + \sqrt{8D_{\text{KL}}(P^* \parallel Q_{\text{LLM}})A_0} \right) \quad (4)$$

2 Proof of Theorem 2 (Fano-Based Regret Lower Bound)

For the well-specified case $D_{\text{KL}} = 0$:

1. Construct a set of $M = \exp(\log K - I_{\text{LLM}})$ problem instances, each with a distinct optimal action. The instances are constructed such that pairwise KL divergence between any two instances is at most $2R_{\max}^2/|\mathcal{Z}|$.
2. By Fano's inequality, the probability of incorrectly identifying the optimal action satisfies $P_e \geq 1/2$ when $T < \Omega(|\mathcal{Z}|(\log K - I_{\text{LLM}}))$.
3. Each identification error incurs regret at least $\Delta_{\min} = \Omega(R_{\max}/\sqrt{|\mathcal{Z}|})$, as the reward gap between the optimal and second-best action scales inversely with the solution space size in the worst case.
4. Multiplying: $\mathbb{E}[\text{Regret}(T)] \geq P_e \cdot \Delta_{\min} = \Omega(\sqrt{T \cdot (\log K - I_{\text{LLM}})})$.

The lower bound matches the upper bound's leading term (up to constants) when $D_{\text{KL}} = 0$, establishing tightness.

3 LLM Prompt Templates

3.1 Prior Acquisition Prompt

The following template is used to obtain LLM action priors:

You are a cybersecurity expert evaluating attack options.

Situation: {state_description}

Available actions: A. {action_1} B. {action_2} ...

Rate each action on a scale of 1-10 based on how likely it is to be the correct next step. Respond ONLY with a JSON object mapping action letters to scores, like: {"A": 8, "B": 3, ...}

3.2 Example: Crypto State

Situation: RSA challenge. $n=3233$, $e=17$, $c=855$. Small modulus suggests weak key.

Available actions: A. Factor n via factordb B. Wiener attack C. Common modulus attack D. Hastad broadcast E. LLL reduction F. Fermat factorization

4 Action Lists and State Descriptions

The complete set of 18 state descriptions (3 per category $\times$ 6 categories) and their corresponding action lists (6 actions each) are provided in the JSON file `states/illm_states.json` in the accompanying code repository at <https://github.com/engin-zhou/cyber-range-experiments>.

5 Experimental Configuration

5.1 Default Hyperparameters

- Number of actions per state: $K = 10$ (synthetic), $K = 6$ (ILLM measurement)
- Noise parameters (synthetic): $p_g = 0.50$, $p_b = 0.28$ ($\Delta = 0.22$)
- Prior weight: $\lambda = 0.05$
- LLM temperature: $\tau = 0.3$
- Trials per condition: $n \in [200, 500]$

5.2 Code and Data Availability

All experiment scripts, data files, and the LLM Gateway implementation are available at <https://github.com/engin-zhou/cyber-range-experiments>. The repository includes a README.md with step-by-step reproduction instructions.